

Regularization - Numerical Questions

Q1. L1 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.5$.

Original loss = 10.

Calculate the regularized loss using L1 regularization.

Formula: $L_{\text{total}} = L + \lambda \sum |w_i|$

Q2. L2 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.1$.

Original loss = 5.

Calculate the regularized loss using L2 regularization.

Formula: $L_{\text{total}} = L + \lambda \sum w_i^2$

Q3. L1 vs L2

Given: $w_1 = 3$, $w_2 = -4$, $w_3 = 2$, $\lambda = 0.2$.

Calculate:

1. L1 penalty
2. L2 penalty
3. L1 regularized loss if original loss = 8
4. L2 regularized loss if original loss = 8

Q4. Ridge Regression

Given: $w_1 = 4$, $w_2 = 3$.

MSE = 6, $\lambda = 0.5$.

Calculate the Ridge regularized loss.

Q5. Lasso Regression

Given: $w_1 = 5$, $w_2 = -2$, $w_3 = 1$.

Loss = 8, $\lambda = 0.4$.

Calculate the Lasso regularized loss.

Q6. L2 Weight Update

Given: $w = 5$, $\eta = 0.1$, $\partial L / \partial w = 2$, $\lambda = 0.5$.

Calculate the new weight using:

$$w_{\text{new}} = w - \eta(\partial L / \partial w + 2\lambda w)$$

Q7. L1 Weight Update

Given: $w = 4$, $\eta = 0.1$, $\partial L / \partial w = 3$, $\lambda = 0.5$.

Calculate the new weight using L1 regularization.

Q8. Ridge with Multiple Weights

Given: $w_1 = 2$, $w_2 = -5$, $w_3 = 3$.

$MSE = 12$, $\lambda = 0.2$.

Calculate the total Ridge loss.

Q9. Lasso with Multiple Weights

Given: $w_1 = -4$, $w_2 = 6$, $w_3 = -2$.

$MSE = 10$, $\lambda = 0.3$.

Calculate the total Lasso loss.

Q10. Elastic Net

Given: $w_1 = 2$, $w_2 = -3$.

$MSE = 7$, $\lambda = 0.5$, $\alpha = 0.6$.

Calculate the Elastic Net regularized loss using:

$$L_{\text{total}} = L + \lambda[\alpha \sum |w_i| + (1 - \alpha) \sum w_i^2]$$